POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

☆ ls ⇒ List all directory & file

☆ ls -l ⇒ ↓ short command ⇒ "ll"

Type	# of link	Owner	Group	Size	Month	Day	Time	Name
drw-r	21	Root	Root	4096	Jan	27	13:3	usr
drwx-xwr	1	"	"	1	Feb	2	14:0	bin
-rw-r	1	"	"	0	May	9	1:2	testfile

☆ (d)rw-xwr ⇒ directory

☆ (l)rwx-xbr ⇒ link

☆ (-)rw-r-l ⇒ Regular file

☆ whoami ⇒ Give user name

☆ pwd ⇒ print working directory

☆ Creating file ⇒ touch
⇒ cp
⇒ vi

☆ Creating directory ⇒ mkdir

Main Ideas, Questions & Summary:

Library / Website Ref.:-

★ touch ⇒ Create a empty file

⇒ touch mayur

⇒ touch testing-file

⇒ if file exist; just timestamp change, everthing remain same.

★ ls -ltr ⇒ List all files/directories according to time in reverse order (older file first).

⇒ Modify files comes at last.

★ cp ⇒ Creates "file" as a copy of another "file".

~~⇒ cp newfile~~ ⇒ Creates file in another location also.

⇒ cp "oldfile" "newfile"

⇒ Creates a "newfile" with a copy of "oldfile".

★ Creating multiple file using touch ⇒

↳ touch "file1" "file2" "file3"

★ mkdir ⇒ Creates a directory

⇒ mkdir "newfolder"

⇒ Also create multiple directory like file.

★ ls -l "filename" ⇒ give details about specific file only.

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

☆ File maintenance commands ⇒

↳ cp
↳ rm
↳ mv
↳ mkdir
↳ rmdir or rm -r
↳ chgrp
↳ chown

☆ mv ⇒ Rename file/directory
⇒ Move files/directory from one location to another.
⇒ mv mayur Raj ⇒ Rename mayur with Raj.
⇒ mv Raj /tmp ⇒ move Raj file to temp directory.
⇒ If folder not exist, give error.

☆ rm ⇒ Used to delete file permanently.
⇒ rm "filename"

Main Ideas, Questions & Summary:

Library / Website Ref.:-

☆ mkdir ⇒ Use to make directory

⇒ mkdir Ram

⇒ mkdir a b c ⇒ Make multiple directory

⇒ If directory exist; does not delete or overwrite it. Give error.

☆ rmdir ⇒ Delete folder only if it is empty.

⇒ rmdir Ram

☆ rm -r ⇒ Delete ~~file~~ folder if not empty also.

⇒ rm -r Ram

☆ chgrp ⇒ Change group ownership of file/folder

↓ Sudo used?

⇒ chgrp root Mayur

GroupName file name

⇒ chgrp -R superhero spiderman ⇒ Changes the grp ownership of file/folder recursively within a specific directory.

☆ chown ⇒ Change Owner & group of file/folder.

↓ Sudo used?

⇒ chown mayur file ⇒ mayur become owner

⇒ chown user:group file ⇒ Change user & group both.

↓ Owner

⇒ chown :group file ⇒ Change group only.

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

✱ Soft & Hard links ⇒

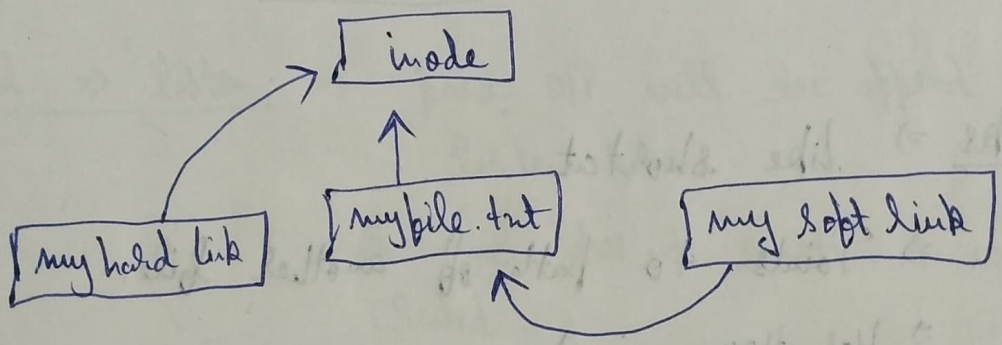
✱ inode ⇒ Pointer / number of a file on hard disk.

✱ Soft links ⇒ Link will be removed if file removed or renamed.

⇒ ln -s

✱ Hard links ⇒ Deleting, renaming or moving the orginal file will not affect the hard link.

⇒ ln



Main Ideas, Questions & Summary:

Library / Website Ref.:-

☆ ls -li ⇒ list inode numbers

☆ Hard link ⇒ Like giving another name to same file.

⇒ Both file points to same inode

⇒ If original file delete, hard link still work

⇒ Because data still exist in inode

Run where you want to create link → ⇒ ln products home/user/items

link name — link of a file

⇒ If modify one file, both change.

⇒ Cannot link directories

⇒ Disk partition should same

☆ Soft links ⇒ like shortcut

⇒ Points to path of another file.

⇒ Has diff inode

⇒ if original file deleted, link become broken.

⇒ ln -s products home/user/items

⇒ Can link directories

⇒ link across different partitions

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ I/P & O/P Redirects ⇒

⇒ 3 Redirects in linux ⇒

① Standard I/P (Stdin) & file descriptor no. as 0 (<, <<)

② Standard O/P (Stdout) & " " " " 1 (>, >>)

③ Standard error (Stderr) & " " " " 2

★ < & << & > & >> ⇒

★ ls -l > listen ⇒ ls -l O/P will get store in listen file & overwrite it.

★ pwd >> listen ⇒ pwd O/P will be append in listen file.

⇒ If file doesn't exist, it will be created.

★ ls -l /root 2 > errorcheck ⇒ If error occur, it will save in file "errorcheck" else run normally.

↙ necessary to use for error.

Main Ideas, Questions & Summary:

Library / Website Ref.:-

★ Pipes ⇒ Used by shell to connect the O/P of one command directly to the I/P of another command.

⇒ denoted by (|).

⇒ ls -l | grep etc

⇒ ls -ltr | more

★ Help Commands ⇒ whatis Command
⇒ command --help
⇒ man Command

★ Linux file editor ⇒ It is a program which enables you to create & manipulate the data (text) in linux file.

⇒ vi ⇒ Visual editor (Run in every linux)
ed ⇒ Standard line editor
ex ⇒ Extended line editor
emacs ⇒ Full screen editor
pico ⇒ Beginner's editor
vim ⇒ Advance version of vi

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ Most common keys in vi =>

i = insert

esc = escape out of any mode

r = replace

d = delete

:q! = quit without saving

:wq! = quit and save (Shift + Z + Z) - shortcut

=> In normal mode (no edit mode) press d+d to delete a line.

=> u to undo last deleted line.

=> /word to search word in a file (also use grep)

Main Ideas, Questions & Summary:

Library / Website Ref.:-

✲ User Account management ⇒

⇒ Commands ⇒ useradd
⇒ groupadd
⇒ userdel
⇒ groupdel
⇒ usermod

} Become root to use these or sudo.

⇒ Files related are ⇒ /etc/passwd (give user info)
⇒ /etc/group (give group info)
⇒ /etc/shadow (give password info)

⇒ id spiderman ⇒ shows user identity & group information.

⇒ gives uid ⇒ User ID
gid ⇒ Group ID
Groups ⇒ All groups user belong to.

✲ userdel ⇒ userdel spiderman ⇒ delete user but user files & user home directory remain.

⇒ userdel -r spiderman ⇒ delete user with files also
↳ recursively.

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ usermod ⇒ modify/change setting of an existing user.

⇒ usermod -g superhero spiderman ⇒ primary group become superhero

⇒ usermod -aG superwoman, spiderman ⇒ Add superwoman as a secondary group & append existing

⇒ usermod -G superhero spiderman ⇒ Replace all secondry groups to superhero only.

★★ useradd -g superhelos -s /bin/bash -c "Ironman" -m -d /home/ironman Ironman

-g: Set primary group
-s: Set shell
-c: Description
-m: Create Home directory
-d: Set path of home directory
Ironman: username

★ passwd ironman ⇒ Set password for ironman user

Main Ideas, Questions & Summary:

Library / Website Ref.:-

☆ /etc/login.defs ⇒ Configuration file that control default setting for new user accounts created using useradd.

⇒ Contain password policies, UID/GID range, home directory setting & other details.

⇒ Contain; PASS_MAX_DAYS ⇒ 99999
PASS_MIN_DAYS ⇒ 0
" _MIN_LEN ⇒ 5
" _WARN_AGE ⇒ 7
& many more ----

⇒ It sets default requirements for all new ~~default~~ users.

☆ chage ⇒ change password setting for a specific user.

⇒ chage -d lastday, -m mindays, -M maxdays, -W warndays,
-I inactive, -E expiredate, username

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ Switch user & sudo access ⇒

⇒ Commands ⇒ su - username

⇒ sudo command

⇒ su (-) → subshell

⇒ visudo (use for editing sudoers file)

⇒ File ⇒ /etc/sudoers ⇒ Control who can use sudo, & what commands they are allowed to run.

⇒ Main security & permission file for sudo.

⇒ wheel group in file (sudoers)

⇒ If user want to run sudo command, then user must be in group named "wheel".

⇒ If user have permission to run all command like root, then it is not necessary for user to be in "wheel" group.

Main Ideas, Questions & Summary:

Library / Website Ref.:-

★ File permission ⇒

⇒ 3 types of permission ⇒ r → read
⇒ w → write
⇒ x → execute (running a program)

⇒ Each permission (rwx) can controlled at 3 levels ⇒

⇒ u → user
g → group
o → other

⇒ chmod ⇒ change permission.

★ chmod g-w jerry ⇒ Retrive write permission from group.

★ chmod a+r jerry ⇒ Give read permission to all (u, g, o)

★ x in file & directory ⇒

⇒ file ⇒ x permission means, file can run as program/script.

⇒ Directory ⇒ You are allowed to enter in directory
⇒ You can access file inside it
⇒ You can use "cd directory"

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

☆ Processes & Job ⇒ Application = Service
⇒ Script
⇒ Process
⇒ Daemon
⇒ Threads
⇒ Job

☆ Monitor & manage Processes command ⇒ df
⇒ du
⇒ uptime
⇒ top
⇒ free
⇒ lsof
⇒ tcpdump
⇒ ps
⇒ kill

(disk free)
↑
☆ df ⇒ Shows how much disk space is used & how much is available on all mounted filesystem.

⇒ df -h ⇒ human readable format

⇒ df -T ⇒ show filesystem type (xfs, ext4)

⇒ df -h /home ⇒ Show specific filesystem

Main Ideas, Questions & Summary:

Library / Website Ref.:-

* du ⇒ disk usage
  ⇒ Shows how much space is being used by files & directories.

⇒ du -h ⇒ human readable

⇒ du -h /home ⇒ Size of home & subdirectories also

⇒ du -sh /home ⇒ Size of home only

* uptime ⇒ Shows ⇒ no of user logged in
  ⇒ How long system has been running
  ⇒ System load averages

* top ⇒ Shows real time information about
  ⇒ CPU usage
  ⇒ Memory usage
  ⇒ Runing processes
  ⇒ Load average
  ⇒ Tasks
  ⇒ System uptime
  ⇒ PID, etc

* free ⇒ Check RAM & swap usage

* lsof ⇒ list open files
  ⇒ In linux, everything is file including directories also.

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☆ kill ⇒ terminate a process

⇒ kill 9982 78

⇒ kill -9 9982813 ⇒ forcefully kill a process if normal kill not work.

☆ tcpdump ⇒ Packet analyzer / n/w sniffer

⇒ Captures & display n/w packets going in/out of your system

⇒ tcpdump -i eth2 ⇒ captures packets on a specific interface

☆ ps ⇒ Shows information about running processes on your system.

⇒ Shows all processes (top shows mostly usage process)

⇒ Show process snapshot (not real time/ live)

⇒ top show realtime / live

⇒ ps -e ⇒ Show all running processes including daemons.

Main Ideas, Questions & Summary:

Library / Website Ref.:-

☆ daemons ⇒ It is a background process that runs automatically without direct user interaction.

⇒ usually end with "d".

⇒ Ex ⇒ sshd

⇒ cloud

☆ Systemd ⇒ It is the first process that starts when linux boots & it's PID is always 1.

⇒ It manages ⇒ System boot process

⇒ Services (daemons)

⇒ Devices

⇒ Logs

⇒ Timer

⇒ network, etc - - -

☆ Systemctl ⇒ It is a command line tool used to interact with Systemd.

⇒ Use to start, stop, enable, disable, check status of services.

⇒ Systemd ⇒ System manager

Systemctl ⇒ Command line tool

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ systemctl --version ⇒ check if systemd installed on system.

★ ps -ef | grep system ⇒ Check if systemd is running

★ systemctl --all ⇒ Check all running services

★ systemctl status | start | stop | restart application.service ⇒

To check status, start, stop, restart a service

★ systemctl reload application.service ⇒

To reload configuration of service

★ systemctl enable | disable application.service ⇒

⇒ To enable or disable service at boot time
⇒ If A require B service, then B start even if it is disable.

★ systemctl mask | unmask application.service ⇒

⇒ To enable/disable service completely (dependency another service dependency)
⇒ If service A require service B to function, then B not start.

★ To stop service use "systemctl stop" not "kill" as it is bad practice.

Main Ideas, Questions & Summary:

Library / Website Ref.:-

☆ SSH ⇒ Stands for Secure Shell

⇒ Open SSH is package / software

⇒ Its service daemon is sshd.

⇒ SSH port #22

⇒ It is a secure n/w protocol used to ⇒

- Remotely login to another linux system
- Execute commands on remote server
- Transfer file securely
- SSH uses encryption, so data is secure

⇒ SSH is itself secure, but there are some additional configuration can be done to make it more secure.

① Configure Idle timeout interval ⇒

- Become root
- Open /etc/ssh/sshd_config file & add following line
  - ⇒ ClientAliveInterval 600
  - ⇒ ClientAliveCountMax 0
- Systemctl restart/reload sshd

⇒ Once the 600 sec interval has passed, idle user will be automatically logged out.

⇒ Always make a copy of original configuration & then edit. because it is best practice & if any error occur we are safe.

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Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

(2) Disable root login ⇒ It disable any user to login to system with root account.

- Become root.
- Edit /etc/ssh/sshd_config file & replace PermitRootLogin Yes to no.
- PermitRootLogin no
- systemctl restart sshd

(3) Disable empty passwords ⇒ It will not allow any user account to login if that account has no password.

- Become root
- Open /etc/ssh/sshd_config file
  - ⇒ Remove "#" from PermitEmptyPasswords no
- systemctl restart sshd

Main Ideas, Questions & Summary:

Library / Website Ref.:-

(4) Limit User SSH Access ⇒ Restrict which users or groups are allowed to log in through SSH.

- Become root
- Open /etc/ssh/sshd_config & add
  - ⇒ AllowUsers mayur user1
  - ⇒ DenyUsers user2
  - ⇒ AllowGroups alex ram wheel
  - ⇒ DenyGroups Tai

  } We can use any one or multiple
- systemctl restart sshd

(5) Use a different port ⇒ By default SSH port runs on 22. Changing it, make system more secure.

- Become root
- Open /etc/ssh/sshd_config file
  - ⇒ remove "#" from "Port 22" & change number.
- systemctl restart sshd.

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Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

☆ Log monitoring ⇒ Process of continously observing & analyzing system & application logs to detect errors, security issues, performance problems.

⇒ Log directory ⇒ /var/log

⇒ Common log files ⇒

- /var/log/messages ⇒ General system msg
- " " /secure ⇒ Authentication & security logs
- " " /boot.log ⇒ Boot related msg
- chronyd ⇒ Synchronize system time
- ~~Cron~~ cron ⇒ time based job scheduler
- maillog
- httpd

⇒ tail -f ⇒ Continously display new lines added to file.
⇒ Show file live updates.

⇒ tail -f "filename"

Main Ideas, Questions & Summary:

Library / Website Ref.:-